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Citation: *The Journal of Chemical Physics* **95**, 628 (1991); doi: 10.1063/1.461413

View online: <http://dx.doi.org/10.1063/1.461413>

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Conformational statistics of semiflexible polymers with excluded volume

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(Received 14 January 1991; accepted 26 March 1991)

The renormalization group treatment (including the crossover effects) of conformational properties of semiflexible polymers with excluded volume is developed in close analogy with that known for the fully flexible case. Obtained results represent further extension of recently discovered connection between Dirac's fermions and semiflexible polymers [Ann. Phys. (N.Y.) **202**, 186 (1990)].

I. INTRODUCTION

It is believed that the rigidity of the polymer chain serves only as some cutoff scale beyond which it can be practically ignored so that polymer chain can be considered as fully flexible.¹ This point of view turns out to be quite successful for a rather wide class of polymer solutions so that a very satisfactory agreement between the theory and experiment has been reached for both static² and dynamic³ problems related to physical properties of polymer solutions. In spite of such success, there are other classes of polymers where rigidity effects play a crucial role. For such classes of polymers well established scaling concepts,¹ as well as methods of renormalization group analysis,² are not applicable directly.

Here we would like to mention several classes of polymers which require explicitly account of flexibility. First, it is useful to distinguish between solutions of neutral polymers, where some sort of liquid crystalline order may exist in the appropriate range of temperatures, and solutions of polyelectrolytes (i.e., polymers which have charges on their backbones) which may undergo conformational changes and changes in liquid crystalline order for some *fixed* temperature when the ionic strength of solution is being changed. Second, the neutral polymer solutions can be further subdivided into solutions of semiflexible homopolymers such as: poly (*p*-phenylene), poly (*p*-phenylene benzobisthiazole), poly (*p*-phenylene benzimidazole), etc.,⁴ and solutions of random copolymers⁵ which contain some random sequence of semiflexible and fully flexible units. "These molecules have varied levels of internal flexibility *but never enough to permit random coil formation*."⁴ Solutions of random semiflexible copolymers exhibit rather interesting phase diagrams⁵ to be discussed in a separate publication. Besides these two classes, there are classes of semiflexible polymers (both random and nonrandom) where rigidity is achieved by replacing polymer's backbone hydrogen by some other chemical groups. As an example,⁴ we would like to mention polyvinyl chloride. The above list of semiflexible polymers can be enlarged by solutions of block copolymers which contain polymers of significantly different flexibility, as well as polymer blends composed of above types of macromolecules, etc. Third, in case of polyelectrolytes the situation is even more interesting. Here rigidity effects can be regulated both chemically and by varying the ionic strength of solution. Unlike the

neutral case, here the situation is considerably complicated by the fact that we must take into account the degree of ionization of the polymer's backbone.⁶ That is, even if we know *a priori* the number and the distribution of charges on the polymer's backbone *before* the polymer was immersed into the solvent, the actual distribution of activated ionized groups on the backbone, once it is in solution, is not known exactly. This fact, when applied even to a single polyelectrolyte chain in solution, makes its description highly nontrivial and requires us to apply for this case the theory of random copolymers.⁷ Besides, "unlike uncharged polymers, configurational properties of polyelectrolytes are not described by simple scaling laws over a wide range of solution conditions. Instead, polyelectrolytes may assume *any shape between that of random coil or rigid rod*, depending on the system's length scales."⁶

Given examples should convince the reader, that the theory of semiflexible polymers deserves as much attention as already well developed theory of fully flexible polymers. Recently, we have made significant progress towards a better understanding of solutions of semiflexible polymers. In Refs. 8 and 9 we have demonstrated that all conformational properties of a single semiflexible macromolecule without excluded volume effects can be described with help of Euclidean version of Dirac's propagator. These results were subsequently generalized to the case of concentrated polymer solutions which resulted in obtaining Flory's⁸ and Onsager's⁹ results for solutions of semiflexible polymers at the lowest order of approximation to the corresponding path integrals. These results were further generalized for the case of random semiflexible copolymers. In Ref. 10 conformational statistics of a single random semiflexible copolymer in the absence of excluded volume was considered. As will become evident from the present work, the excluded volume effects for semiflexible random copolymers can be treated almost in the same way as for the fully flexible case.²

Recently, we have considered conformational properties of polyelectrolytes using various methods. In Ref. 11 the renormalization group method was used. This method turned out to be of limited value because it was limited only to highly screened chains for which electrostatic interactions are very weak. When the strength of these interactions is not small, other methods were used¹² culminating in the variational-type calculations¹³ similar to that used for Feynman's large polaron problem.¹⁴ We were able

to map the electrostatic problem into that of the *neutral* semiflexible polymer chain. For this chain the rigidity parameter was variationally calculated as some *known* function of the ionic strength of solution for the fixed temperature. Hence, the problems related to conformational properties of dilute polyelectrolyte solutions with excluded volume can be mapped into that for the neutral semiflexible polymers. This point of view will be further examined in the present paper.

This paper is organized as follows. In Sec. II we provide the necessary background material related to both flexible and semiflexible polymers. Here some aspects of field-theoretic treatments of fully flexible polymers are revised in order to be further extended for the semiflexible case discussed in Sec. III. Section IV is devoted to some illustrative calculations such as computations of the reduced free energy and the mean square end-to-end distance $\langle R^2 \rangle$. Section V is devoted to the detailed crossover calculations based on our earlier developed results¹⁵ for the fully flexible polymers. Finally, Sec. VI is devoted to a brief summary and discussion while the Appendix serves to provide some auxiliary technical details needed in the main text.

II. STATEMENT OF THE PROBLEM

It is well known¹ that the ordinary random walk in d dimensions can be fully described in terms of the generating function $G_0(\mathbf{p}, N)$ given by

$$G_0(\mathbf{p}, N) = \exp \left\{ -\frac{Nl^2}{2d} \mathbf{p}^2 \right\}, \quad (2.1)$$

where N is the number of steps in the walk and l is order of the lattice spacing (which is chosen to be the same as the persistent length^{1,2} if the continuum description is used). Equation (2.1) allows us to obtain the important information about the walk which is given by the moments. For example, the mean square end-to-end distance is given by

$$\langle R^2 \rangle = (-2d) \frac{d}{d\mathbf{p}^2} \ln G_0(\mathbf{p}, N) \Big|_{\mathbf{p}=0} = Nl^2, \quad (2.2)$$

etc.

For many purposes it is more convenient to consider along with $G_0(\mathbf{p}, N)$ its Laplace transform given by

$$G_0(\mathbf{p}, s) = \left(\frac{l^2}{2d} \mathbf{p}^2 + s \right)^{-1}, \quad (2.3)$$

where s is Laplace variable conjugate to N . It is convenient to rescale s variable $s \rightarrow s'(l^2/d)$ so that Eq. (2.3) can be rewritten as

$$G_0(\mathbf{p}, s) = \frac{d}{l^2} \left(\frac{\mathbf{p}^2}{2} + s \right)^{-1}. \quad (2.4)$$

The factor d/l^2 can be absorbed into $G_0(\mathbf{p}, s)$ which is going to be assumed below. Equation (2.4) also can be obtained from the path integral as it is described in Ref. 2. For completeness, we provide here some details.

In the absence of effects of excluded volume, fully flexible polymers can be modeled by the random walks. The end-to-end distribution function for such walks is given by²

$$G_0(\mathbf{R}, N) = \int_{\mathbf{r}(0)=0}^{\mathbf{r}(N)=\mathbf{R}} \mathcal{D}[\mathbf{r}(\tau)] \exp \left\{ -\frac{d}{2l} \int_0^N d\tau \left(\frac{d\mathbf{r}}{d\tau} \right)^2 \right\}. \quad (2.5)$$

Here $\mathbf{r}(\tau)$ represents a spatial position of the polymer segment which has contour position τ along the polymer chain. Calculation of the propagator given by Eq. (2.5) proceeds in complete analogy with that for nonrelativistic free particle. Fourier transforming of $G_0(\mathbf{R}, N)$ produces back Eq. (2.1). By analogy with quantum mechanics,¹⁴ it can be easily demonstrated, that $G_0(\mathbf{R}, N)$ can be obtained as solution of the diffusion equation given by

$$\left(\frac{\partial}{\partial N} - \frac{l^2}{2d} \nabla^2 \right) G_0(\mathbf{R}, N) = \delta(\mathbf{R}) \delta(N). \quad (2.6)$$

Fourier-Laplace transforming of Eq. (2.6) brings us back to Eq. (2.3) as required. Equation (2.5) is convenient for developing the theory of random walks. For example, if we impose the extra restriction that the walk is not allowed to intersect itself, then for such self-avoiding walks (SAW) we can write the following distribution function:

$$G(\mathbf{R}, N) = \int_{\mathbf{r}(0)=0}^{\mathbf{r}(N)=\mathbf{R}} \mathcal{D}[\mathbf{r}(\tau)] \exp \{ -H[\mathbf{r}(\tau)] \}, \quad (2.7)$$

where

$$H[\mathbf{r}, \tau] = \frac{d}{2l} \int_0^N d\tau \left(\frac{d\mathbf{r}}{d\tau} \right)^2 + \frac{\mathcal{V}}{2} \int_0^N d\tau \int_0^N d\tau' \delta[\mathbf{r}(\tau) - \mathbf{r}(\tau')] \quad (2.8)$$

and in the second term in Eq. (2.8) it is implicitly assumed that the excluded volume interactions, characterized by the coupling constant $\mathcal{V}$, cannot take place for the contour distances $|\tau - \tau'| \leq L$ with L being some cut off of order l (to be discussed in detail below). When Eq. (2.8) is substituted into Eq. (2.7) the perturbation expansion of $G(\mathbf{R}, N)$ can be obtained in powers of excluded volume $\mathcal{V}$. This procedure is described in detail in Refs. 2 and 16. Fourier-Laplace transforming of such perturbation expansion allows us to write the following compact expression for $G(\mathbf{p}, s)$ in the presence of excluded volume:

$$[G(\mathbf{p}, s)]^{-1} = \frac{\mathbf{p}^2}{2} + s + \Sigma(\mathbf{p}, s). \quad (2.9)$$

Here $\Sigma(\mathbf{p}, s)$ accounts for all excluded volume effects and can be calculated perturbatively order by order in $\mathcal{V}$, as it is described in Ref. 16. Equation (2.9) is meaningless as it is written because of divergences arising in computation of $\Sigma(\mathbf{p}, s)$. The systematic procedure of removal of the divergences was developed in Ref. 16 and the interested reader is encouraged to consult the above reference. After the removal of singularities, the regularized expression for

$G(\mathbf{p}, s)$ cannot be uniquely defined which is very convenient for the development of the renormalization group methods.^{2,16} These methods, as is well known, are targeted towards obtaining of large scale asymptotic information about the system which, in the case of polymers, is given by various power law dependencies,¹ crossover effects,¹⁵ etc.

It is possible to extend these results for the case of semiflexible polymers, as it was, in part, accomplished already in Refs. 9 and 10. Here we provide some important additional details. The development can be accomplished in several ways. For instance, we can rewrite Eq. (2.9) as follows:

$$G(\mathbf{p}, s) = G_0(\mathbf{p}, s) - G_0(\mathbf{p}, s) \sum (\mathbf{p}, s) G_0(\mathbf{p}, s) + G_0(\mathbf{p}, s) \times \left[\sum (\mathbf{p}, s) G_0(\mathbf{p}, s) \right]^2 + \dots, \quad (2.10)$$

where in the case of flexible polymers $G_0(\mathbf{p}, s)$ is given by Eq. (2.4). If the semiflexible polymers could be described (in the absence of excluded volume effects) by some propagator $\hat{G}_0(\mathbf{p}, s)$ which, evidently should differ from G_0 given by Eq. (2.4), then the excluded volume effects can be accounted for by formally replacing $G_0(\mathbf{p}, s)$ by $\hat{G}_0(\mathbf{p}, s)$ in Eq. (2.10). This is permissible by the principle of consistency: At elevated enough temperatures, semiflexible polymers are indistinguishable from the usual flexible polymers.^{1,17} Moreover, the flexible polymers *must* be considered only as a limiting case of the semiflexible polymers and not the other way around. The above formal replacement of G_0 by $\hat{G}_0$ is especially convenient if the exact path integral representation of $\hat{G}_0$ is unknown or rather cumbersome. Moreover, for almost all (*but not all*) practical purposes there is no need in knowing the corresponding path integral for $\hat{G}_0$, which in the above limiting case is expected to be reducible to $G_0(\mathbf{R}, N)$ given by Eq. (2.5). Therefore, in this paper we do not provide the corresponding path integral expression for $\hat{G}_0$. The interested reader can find some information in this regard by reading our earlier work, Ref. 9, and references therein.

Although the above replacement is rather plausible as far as the actual calculations are concerned, it leaves some issues without explanation. For example: (a) how the above perturbational expansion can be renormalized; (b) how the renormalization group methods can be used if, apparently, semiflexible polymers in most of the instances do not exhibit the usual scaling behavior, as it was discussed in Sec. I, etc. The most convenient way to study the renormalization is to convert the polymer problem into the field-theoretic form.¹⁸ For completeness we present here a brief summary of this procedure. Additional details can be found in the cited literature.¹⁸

In view of Eqs. (2.7) and (2.8), consider the following Hubbard–Statonovich identity:

$$\exp \left\{ -\frac{1}{2} \int_0^N d\tau \int_0^N d\tau' W[\mathbf{r}(\tau) - \mathbf{r}(\tau')] \right\} = \mathcal{N} \int \mathcal{D}[\varphi] \exp \left\{ -\frac{1}{2} \int d\mathbf{r} \int d\mathbf{r}' \varphi(\mathbf{r}') \right.$$

$$\times W^{-1}(\mathbf{r}' - \mathbf{r}) \varphi(\mathbf{r}) \pm i \int_0^N d\tau \varphi[\mathbf{r}(\tau)] \left. \right\}. \quad (2.11)$$

Here W is some well defined binary potential (in our case it is a delta function) which has an inverse and the normalization constant $\mathcal{N}$ is chosen in such a way, that when the square is completed in the right-hand side of Eq. (2.11) the left-hand side is reproduced. Combining Eq. (2.11) with Eqs. (2.7) and (2.8) allows us to rewrite $G(\mathbf{R}, N)$ in the following equivalent form:

$$G(\mathbf{R}, N) = \langle G(\mathbf{R}, N; \varphi) \rangle_\varphi, \quad (2.12)$$

where $\langle \dots \rangle_\varphi$ is given by

$$\langle \dots \rangle_\varphi = \mathcal{N} \int \mathcal{D}[\varphi] \exp \left\{ -\frac{1}{2\mathcal{V}} \int d\mathbf{r} \varphi(\mathbf{r}^1) W^{-1} \times (\mathbf{r}^1 - \mathbf{r}) \varphi(\mathbf{r}) \right\} \dots, \quad (2.13)$$

and $G(\mathbf{R}, N; \varphi)$ satisfy the following equation of “motion”:

$$\left[\frac{\partial}{\partial N} - \frac{l^2}{2d} \nabla^2 \pm i\varphi(\mathbf{r}) \right] G(\mathbf{R}, N; \varphi) = \delta(\mathbf{R}) \delta(N), \quad (2.14)$$

to be compared with Eq. (2.6). Use of Laplace transform converts Eq. (2.14) into

$$\left[s - \frac{l^2}{2d} \nabla^2 \pm i\varphi(\mathbf{r}) \right] G(\mathbf{R}, s; \varphi) = \delta(\mathbf{R}). \quad (2.15)$$

Function $G(\mathbf{R}, s; \varphi)$ can be equivalently represented with help of another identity as follows:

$$G(\mathbf{R}, s; \varphi) = Z^{-1} \int \mathcal{D}[\Psi] \Psi(\mathbf{R}) \Psi(0) \exp \left\{ -\frac{1}{2} S_0[\varphi, \Psi] \right\}, \quad (2.16)$$

where

$$S_0[\varphi, \Psi] = \int d\mathbf{r} \left[\frac{l^2}{2d} (\nabla \Psi)^2 + s \Psi^2 + i\varphi \Psi^2 \right] \quad (2.17)$$

and

$$Z = \int \mathcal{D}[\Psi] \exp \left\{ -\frac{1}{2} S_0[\varphi, \Psi] \right\}. \quad (2.18)$$

Combining Eqs. (2.12), (2.13), and (2.16)–(2.18) and performing the trivial Gaussian integration over φ field (in expense of introducing replicas) we obtain¹⁸

$$G(\mathbf{R}, s) = \lim_{n \rightarrow 0} \sum_{\alpha=1}^n \frac{\mathcal{N}}{n} \int \mathcal{D}[\Psi] \Psi_\alpha(\mathbf{R}) \Psi_\alpha(0) \times \exp \{ -S[\Psi] \} \quad (2.19)$$

where

$$\mathcal{D}[\Psi] = \prod_{\alpha=1}^n \mathcal{D}[\Psi_\alpha],$$

$$S[\Psi] = \frac{1}{2} \int d\mathbf{r} \left[\frac{1}{2} (\nabla \Psi)^2 + s \Psi^2 + \frac{g_0}{4} (\Psi^2)^2 \right], \quad (2.20)$$

$$(\nabla; \Psi) = \sum_{\alpha=1}^n \nabla \Psi_{\alpha} \Psi^2 = \sum_{\alpha=1}^n \Psi_{\alpha} \Psi_{\alpha},$$

and the variable s [see Eq. (2.4)], as well as the fields Ψ_{α} , were rescaled so that the factor near the gradient term becomes equal to $1/2$. The coupling constant g_0 is related to $\mathcal{V}$ as $\mathcal{V}(2d/l^2)^2 = g_0$. As in all field theory calculations,¹⁹ the actual value of g_0 remains as a parameter of the theory and, hence, is seemingly unimportant as far as only the scaling behavior is of interest to us. This is *not* going to be the case for semiflexible polymers as we shall demonstrate below. For the time being, it is noteworthy to notice that for the fixed $\mathcal{V}$ when $l^2 \rightarrow \infty, g_0 \rightarrow 0$ and when $l^2 \rightarrow 0, g_0 \rightarrow \infty$. This *already* indicates that the rigidity (given in terms of the persistence length l) plays an important role in crossover effects, i.e., when the polymer chain changes its conformation from that characteristic of good solvent regime to that of the theta regime. It is obvious, that the existing theory of fully flexible chains is appropriate whenever $0 < l \ll Nl$ while for the case of semiflexible polymers we may have $0 < l \ll Nl$ as it will be discussed below.

Above presented results, when adopted for semiflexible polymers, must reduce to the flexible case in the high temperature limit, as was already stated. Hence, our task is to find an appropriate generalization of Eqs. (2.19) and (2.20) so that known results of the field theory can be utilized. This is going to be accomplished in the next section.

III. FIELD THEORY FOR SEMIFLEXIBLE POLYMERS

Begin with the well known result¹⁷ from the theory of semiflexible polymers,

$$\langle R^2 \rangle = 2\hat{N}\eta - 2\eta^2 \left[1 - \exp\left(-\frac{\hat{N}}{\eta}\right) \right], \quad (3.1)$$

where η is phenomenological rigidity parameter. For $\eta \rightarrow \infty$ we obtain $\langle R^2 \rangle = \hat{N}^2$ while in the opposite case, $\eta \rightarrow 0$, we obtain $\langle R^2 \rangle \simeq 2N\eta$. Comparison of these results with Eq. (2.2) indicates that $\hat{N} = Nl$ and $2\eta = l$. It is important to realize the for $\eta = 0$ in Eq. (3.1) the left-hand side loses its meaning so that the above identification indeed makes sense in view of the results presented in Sec. II. It is legitimate to ask now the following question: What is the analog of the generating function, Eq. (2.1), for the case of semiflexible polymers?

Recently, we have demonstrated⁹ that the generating function for the semiflexible polymers is closely associated with the Euclidean version of the Dirac propagator. For completeness, we provide here some necessary information to make our presentation self-contained.

Following Feynman,²⁰ consider a square lattice which is rotated counterclockwise by 45° angle so that the axes of some chosen coordinate system go through the diagonals of the initial lattice squares. Consider the motion of a particle on such rotated lattice. The particle is allowed to go straight or to change its direction by 90° at the corners of squares of the lattice. In the X - Y plane, let X axis represent "space" while Y axis "time". The particle is allowed to move back and forth along the direction of the X axis

but only forth along the Y axis direction. Let $a = (x_1, y_1)$ and $b = (x_2, y_2)$ be the initial and final points for such type of a walk and let $N(R)$ be the total number of walks of N steps long, starting at a , ending at b , and having exactly R turns along their path. The lattice propagator $K_{\alpha\beta}(a, b)$ can be written then in a form

$$K_{\alpha\beta}(a, b) = \sum_{R=0}^{\infty} N_{\alpha\beta}(R) (iem)^R, \quad (3.2)$$

where $\alpha(\beta)$ correspond to the initial (final) orientation of the walk: to the "right" (+) or to the "left" (-). It is being assumed, that the length of each step is ϵ and m represents the *trans-gauche* bending factor: To go straight costs nothing but to change the direction costs m each time. The imaginary i factor is included because initially the problem is studied in Minkowskii space in connection with Dirac's propagator. Usually, we have to make an average over the initial positions and sum over the final ones so that the actual quantity of interest is

$$K(a, b) = \sum_{\alpha, \beta} K_{\alpha\beta}(a, b) p(\alpha), \quad (3.3)$$

where $p(\alpha)$ represents the discrete version of the polarization matrix for Dirac particle.²¹ In the subsequent, only the "nonpolarized" case $p(+)=p(-)=\frac{1}{2}$ is going to be considered and the above factors are being absorbed into $K_{\alpha\beta}$. In case if conformational phase transitions in membranes are of interest, polarization effects are of importance and must be taken explicitly into account. In the present work we are not going to discuss this situation further. With this introduction, we are ready to reformulate our problem into statistical mechanics language. If N_+ (N_-) is the total number of steps to the right (left), then $N = N_+ + N_-$. Introduce as well $M = N_+ - N_-$, then the "time" interval $t_b - t_a$ is given, in view of above definitions, by

$$t_b - t_a = N\epsilon, \quad (3.4)$$

while the space interval is given by

$$x_b - x_a = M\epsilon. \quad (3.5)$$

Introduce now a system of N Ising spins σ_i , $i = 1 - N$, and associate the i th step with σ_i so that $\sigma_i = +1$ (-1) will correspond to the move to the right (left). With such a defined rule for spins, the "magnetization" M is given by

$$M = N_+ - N_- = \sum_{i=1}^N \sigma_i. \quad (3.6)$$

The number of turns can now be easily obtained as

$$R = \frac{1}{2} \sum_{i=1}^{N-1} (1 - \sigma_i \sigma_{i+1}). \quad (3.7)$$

Use of Eq. (3.5), allows us to write for the propagator $K_{\alpha\beta}(a, b)$, the following result:

$$K_{\alpha\beta}(a, b) = Z_{\sigma_1 \sigma_N}(N, M; j) \\ = \sum'_{\sigma_2 = \pm 1} \cdots \sum'_{\sigma_{N-1} = \pm 1} \\ \times \exp \left\{ j \sum_{i=1}^{N-1} (\sigma_i \sigma_{i+1} - 1) \right\}, \quad (3.8)$$

where $j = (-1/2) \ln(i\epsilon m)$ and primes near the summation signs indicate that the partition function Z should actually be evaluated under the constraint given by Eq. (3.6). Explicit account of the above constraint leads to the following "momentum-space" representation of the propagator:

$$Z_{\sigma_1 \sigma_N}(N, \mu; j) = \sum_{M=-N}^N e^{\mu M} Z_{\sigma_1 \sigma_N}(N, M; j) \quad (3.9)$$

and spin summation goes from σ_2 to σ_{N-1} , as before. If $t = t_b - t_a$ and $x = x_b - x_a$, then $Z(N, M; j) \rightarrow K(t, p)$ with momentum variable p being determined by the relation $m\mu = -ipx$, or

$$\mu = -ip\epsilon. \quad (3.10)$$

Use of standard transfer matrix methods produces now

$$Z_{\sigma_1 \sigma_N}(N, \mu; j) = \lambda_+^N \varphi_+(\sigma_1) \varphi_+(\sigma_N) \\ + \lambda_-^N \varphi_-(\sigma_1) \varphi_-(\sigma_N), \quad (3.11)$$

where $\lambda_{\pm}$ and $\varphi_{\pm}$ are eigenvalues and eigenfunctions of the matrix $H(\sigma_b \sigma_{i+1})$ given by

$$H(\sigma_b \sigma_{i+1}) = \exp \left\{ \frac{\mu}{2} (\sigma_i + \sigma_{i+1}) + j (\sigma_i \sigma_{i+1} - 1) \right\}. \quad (3.12)$$

In actual computations it is essential to remember that $\exp(-2j) = i\epsilon m$ and $\mu = -ip\epsilon$. This then produces the matrix

$$H = \begin{pmatrix} 1 - ip\epsilon & i\epsilon m \\ i\epsilon m & 1 + ip\epsilon \end{pmatrix}, \quad (3.13)$$

which has eigenvalues

$$\lambda_{\pm} = 1 \pm i\epsilon m E, \quad (3.14)$$

where

$$E^2 = 1 + \frac{p^2}{m^2}. \quad (3.15)$$

With help of Eq. (3.13) eigenfunctions of the matrix H can be easily obtained thus producing the final result,

$$G(p, t) = \sum_{\alpha, \beta} K_{\alpha\beta}(a, b) \\ = \left(1 - \frac{1}{E}\right) e^{-imEt} + \left(1 + \frac{1}{E}\right) e^{imEt}. \quad (3.16)$$

In arriving at this result it was taken into account that $N = t/\epsilon$ and

$$\lim_{\epsilon \rightarrow 0^+} (1 \pm i\epsilon m E)^{1/\epsilon} = \exp(\pm imEt). \quad (3.17)$$

If a multidimensional situation is being considered, then we have to replace p^2 by $\mathbf{p}^2$ as can be seen easily from the nonrelativistic limit ($\mathbf{p}^2/m^2 \ll 1$) of the above propagator. In this limit $E \simeq 1 + \mathbf{p}^2/2m^2$ so that using Eq. (3.16) we arrive at the familiar result given in Feynman's book:²¹

$$G(\mathbf{p}, t) \simeq 2e^{imt} e^{i\mathbf{p}^2 t/2m}. \quad (3.18)$$

Apart from the unimportant normalization factor $2e^{imt}$, this result coincides with the result for the propagator for the nonrelativistic pointlike free particle which has mass m . In Sec. II we have already mentioned that this propagator coincides, up to normalization, with that for the random walk problem, Eq. (2.5). Specifically, if the Euclidean rotation $it \rightarrow -\hat{N} \equiv Nl$ is performed in Eq. (3.18) and $m = d/l$, then we would obtain back Eq. (2.1) for the generating function. From the above identification, we notice that the mass parameter in Eq. (3.16) is related to the persistence length. Moreover, as we have demonstrated before,⁹ the mass m is directly related to the *trans-gauche* energy ΔE via relationship $\epsilon m = (z - 2) \exp(-\Delta E/T)$, where z is the coordination number of the lattice (if we initially consider the case of packing polymers on the lattice). We would like now to demonstrate that the phenomenological rigidity parameter η is also directly connected with m . To this purpose we perform the Euclidean rotation $it \rightarrow \hat{N}$ and put $\mathbf{p} \Rightarrow (i\mathbf{p}/2d)$ in Eq. (3.16), then we obtain

$$\hat{G}_0(\mathbf{p}, \hat{N}) = 2 \cosh(mE\hat{N}) + \frac{2}{E} \sinh(mE\hat{N}). \quad (3.19)$$

$\langle \mathbf{R}^2 \rangle$ can now be obtained by using the formula Eq. (2.2) with G_0 being replaced by $\hat{G}_0$ given above. This produces⁹

$$\langle \mathbf{R}^2 \rangle = \frac{(-2d)}{\hat{G}_0} \frac{\partial \hat{G}_0}{\partial E} \frac{\partial E}{\partial \mathbf{p}^2} \Big|_{\mathbf{p}=0} \\ = \frac{\hat{N}}{2m} - \frac{1}{4m^2} (1 - e^{-2m\hat{N}}). \quad (3.20)$$

Comparison between Eqs. (3.1) and (3.20) suggests that $m = (2\sqrt{2}\eta)^{-1}$ if the length $\hat{N}$ is rescaled as $\hat{N} \Rightarrow \hat{N}\sqrt{2}$. The necessity to rescale the length $\hat{N}$ is easy to understand: The vertical time coordinate N and the actual number of steps (of unit length in suitably chosen system of units) are related to each other as $\hat{N} = N\sqrt{2}$ if the underlying lattice is a simple square lattice. Because of the earlier demonstrated relationship between the mass m and the *trans-gauche* energy ΔE , the phenomenological rigidity parameter η is shown to be directly related to the microscopic parameters of polymers which can be computed, in principle, quantum mechanically. In case of polyelectrolytes, as we have demonstrated earlier,¹³ the rigidity parameter η is a function of the ionic strength of solution so that m may also depend on the ionic strength.

As in case of Eq. (2.3), use of Laplace transform applied to Eq. (3.19) produces

$$\hat{G}_r(\mathbf{p}, s) = \frac{2(s + m)}{s^2 - m^2 E^2}, \quad (3.21)$$

where now $E^2 = 1 - (\mathbf{p}^2/2 dm^2)$. Use of this fact allows us to rewrite Eq. (3.21) in the following form:

$$\hat{G}_0(\mathbf{p}, s) = \frac{2d(s+m)}{(\mathbf{p}^2/2) + M^2}, \quad (3.22)$$

where $M^2 = d(s^2 - m^2)$. It is of interest to find the nonrelativistic limit ($m \rightarrow \infty$, $\mathbf{p} \rightarrow 0$) for the above propagator. In this limit we have

$$\begin{aligned} \hat{G}(\mathbf{p}, s) &= \frac{2(s+m)}{(2-mE)(s+mE)} \\ &\approx \frac{2}{s-mE} \left(1 + \frac{\mathbf{p}^2}{4dm(s+m)} \right) \\ &\approx \frac{2}{s-m + (\mathbf{p}^2/2m')} \end{aligned} \quad (3.23)$$

where $m' = 2dm$, to be compared with Eq. (2.3). A formal use of the propagator, Eq. (3.22), in expansion, Eq. (2.10), would lead to perturbation expansion similar to that for the fully flexible polymers. The renormalization of this expansion can, however, be better understood with use of the replica trick formalism. By analogy with Eqs. (2.19) and (2.20) and taking into account that in the nonrelativistic limit, the propagator, Eq. (3.23), practically coincides with that for the fully flexible polymers, Eq. (2.4), we can write now the following result (after obvious rescaling of the field Ψ_α as compared with the fully flexible case):

$$\begin{aligned} \hat{G}(\mathbf{R}, s) &= \lim_{n \rightarrow 0} \sum_{\alpha=1}^n \frac{\mathcal{N}}{n} \int \mathcal{D}[\Psi] \Psi_\alpha(\mathbf{R}) \Psi_\alpha(0) \\ &\quad \times \exp\{-\hat{S}[\Psi]\}, \end{aligned} \quad (3.24)$$

where

$$\begin{aligned} \mathcal{D}[\Psi] &= \prod_{\alpha=1}^n \mathcal{D}[\Psi_\alpha], \\ S[\Psi] &= \frac{1}{2} \int d\mathbf{r} \left[\frac{1}{2} (\nabla \Psi)^2 + M^2 \Psi^2 + \frac{\hat{g}_0}{4} (\Psi^2)^2 \right], \end{aligned} \quad (3.25)$$

and $\hat{g}_0 = \mathcal{V} 4d^2(s+m)^2$.

As long as the inverse Laplace transform of the quantities of interest is not made, the actual value of the coupling constant $\hat{g}_0$ is of no importance as was already noticed in Sec. II. But when the problem of crossover is of interest, there is a significant difference between the cases of fully flexible and semiflexible polymers because of the presence of mass m in the coupling constant. As in standard field-theoretic case,¹⁹ the dimensional analysis of the action $\hat{S}$ can now be performed. If L is some yet arbitrary length scale, then the requirement that the action $\hat{S}$ is dimensionless produce: $[\Psi] = L^{1-(d/2)}$, $[\hat{g}_0] = L^\epsilon$, where $\epsilon = 4-d$, and $[M^2] = L^{-2}$. It is useful to formally introduce the dimensionless coupling constant g , so that $\hat{g}_0 = gL^\epsilon$ in Eq. (3.25). After these redefinitions, the field theory for the semiflexible polymer chain, given by Eqs. (3.24) and (3.25) is almost the same as that for the fully flexible polymers, Eqs. (2.19) and (2.20), except now the mass $M^2 \neq s$ as in the fully flexible case. This makes no

difference as far as formal field-theoretic calculations are performed, but it produces quite significant changes upon the use of the inverse Laplace transform to the obtained results. This is going to be considered in some detail in the next section.

IV. ILLUSTRATIVE CALCULATIONS OF $\langle R^2 \rangle$

In Ref. 16, detailed calculations of $\langle R^2 \rangle$ for the flexible polymer chain with excluded volume were performed. For the reader's convenience, we would like to provide here some key elements of the above calculations in a rather simplified form which are required for the subsequent development. Use of Eqs. (2.4) and (2.10) produces (in the first order in $\mathcal{V}$) the following result for $G(\mathbf{p}, s)$:

$$G(\mathbf{p}, s) \approx G_0(\mathbf{p}, s) - G_0(\mathbf{p}, s) \sum^{(1)}(\mathbf{p}, s) G_0(\mathbf{p}, s) + \dots, \quad (4.1)$$

where the ellipses denote all higher order terms. We have obtained before¹⁶

$$\sum^{(1)}(\mathbf{p}, s) = -\frac{2\mathcal{V}_0 s}{\epsilon} + \mathcal{V}_0 s \left[\ln\left(\frac{sL}{2\pi}\right) - \psi(2) \right], \quad (4.2)$$

where $\psi(2) = 1 - \gamma$ and γ is Euler's constant. The first term is removed by renormalization as is described in detail in Ref. 16. After that, the bare coupling constant $\mathcal{V}_0$ is replaced by the renormalized dimensionless coupling constant $\mathcal{V}$ and Eq. (4.2) can be substituted in (4.1). Use of Eq. (2.4) and choice of the system of units such that $d/l=1$ produces for $G(\mathbf{p}, s)$ the following result:

$$\begin{aligned} G(\mathbf{p}, s) &\approx \frac{1}{(\mathbf{p}^2/2) + s} - \frac{\mathcal{V}s}{[(\mathbf{p}^2/2) + s]^2} \\ &\quad \times \left[\ln\left(\frac{sL}{2\pi}\right) - \psi(2) \right] + \dots \end{aligned} \quad (4.3)$$

This further produces

$$G(0, s) = \frac{1}{s} - \frac{\mathcal{V}}{s} \left[\ln\left(\frac{sL}{2\pi}\right) - \psi(2) \right] + \dots \quad (4.4)$$

Use of the inverse Laplace transform converts the last expression into

$$\begin{aligned} G(0, \hat{N}) &= 1 + \mathcal{V} \left(1 + \ln\left(\frac{2\pi\hat{N}}{L}\right) \right) \\ &\approx (1 + \mathcal{V}) \left(1 + \mathcal{V} \ln\left(\frac{2\pi\hat{N}}{L}\right) \right). \end{aligned} \quad (4.5)$$

The last result can be rewritten as well (for $\mathcal{V}$ small) as

$$G(0, \hat{N}) \approx \left(\frac{2\pi\hat{N}}{L} \right)^{\mathcal{V}} (1 + \mathcal{V}). \quad (4.6)$$

If $\mathcal{V}$ is taken at its fixed point value¹⁶, $\mathcal{V}^* = \epsilon/8$, then the last result can be equivalently rewritten as^{2,16}

$$G(0, \hat{N}) = \left(\frac{2\pi\hat{N}}{L} \right)^{\gamma-1} \text{const}, \quad (4.7)$$

where the “susceptibility” exponent γ is given by $\gamma = 1 + \epsilon/8$ and $\text{const} = 1 + \epsilon/8$. The last result has a clear thermodynamic meaning as explained in Ref. 1. It is related to the ratio of the total number of selfavoiding walks of length $\hat{N}$ to the total number of random walks without the selfavoidance constraint. Because so far the arbitrary length L in Eq. (4.7) was not specified, Eq. (4.7) is only of a limited value.

In case of semiflexible polymers, the role of L is going to be completely determined. *This represents one of the major advantages of the theory of semiflexible polymers as compared to the fully flexible case.* For the semiflexible polymers, instead of the mass operator given by Eq. (4.2), we obtain

$$\Sigma^{(1)}(\mathbf{p}, s) = -\frac{2\mathcal{V}M^2}{\epsilon} + \hat{g}_0 M^2 \left(\ln \left(\frac{M^2 L^2}{2\pi} \right) - \psi(2) \right) \quad (4.8)$$

after renormalization this produces the following result for $\hat{G}(0, s)$:

$$\begin{aligned} \hat{G}(0, s) &= \frac{2d(s+m)}{M^2} - \frac{\mathcal{V}}{M^2} [2d(s+m)]^3 \\ &\times \left[\ln \left(\frac{M^2 L^2}{2\pi} \right) - \psi(2) \right]. \end{aligned} \quad (4.9)$$

In arriving at Eq. (4.9) we have used Eqs. (3.22), (3.25), and (4.1). The last equation can be equivalently rewritten as

$$\begin{aligned} \hat{G}(0, s) &= \frac{2}{s-m} - \frac{\mathcal{V}8d^2(s+m)^2}{s-m} \left[\ln \left(\frac{(s-m)}{2\pi} \sqrt{dL} \right) \right. \\ &\left. + \ln \left(\frac{(s+m)}{2\pi} \sqrt{dL} \right) - \psi(2) \right]. \end{aligned} \quad (4.10)$$

Use of Laplace transform in the present case is somewhat technically involved as compared to the fully flexible case, so that we present all relevant calculations in the Appendix. Here we provide only the final result,

$$\begin{aligned} \hat{G}(0, N) &= 2e^{m\hat{N}} \left\{ 1 + u \left[\ln \left(\frac{2\pi\hat{N}}{\sqrt{dL}} \right) \right. \right. \\ &\left. \left. - \ln \left(\frac{2m\sqrt{d}}{2\pi} L \right) + 1 \right] \right\}, \end{aligned} \quad (4.11)$$

where $u = \mathcal{V}(4md)^2$. Comparison between Eqs. (4.5) and (4.11) leads us to conclude: *in order to reach consistency between the present case and the theory of fully flexible polymers we must require $mL\sqrt{d}/\pi = 1$, or*

$$L = \frac{\pi}{m\sqrt{d}}. \quad (4.12)$$

Hence, in the theory of semiflexible polymers, the arbitrary length scale L present in Eqs. (4.5) and (4.6) is completely fixed by the rigidity of the chain. Moreover, the rigidity directly affects the coupling constant u so that the crossover effects occur mainly because of the changes in the flexibility of the chain's backbone.

In order to compute $\langle \mathbf{R}^2 \rangle$, several more steps are required as explained in Ref. 16. Consider first a fully flexible case. For this case, using Eq. (4.3), we have to calculate

$$-2 \frac{\partial}{\partial \mathbf{p}^2} G(\mathbf{p}, s) \Big|_{\mathbf{p}=0} = \frac{1}{s^2} - \frac{2\mathcal{V}}{s^2} \left[\ln \left(\frac{sL}{2\pi} \right) - \psi(2) \right]. \quad (4.13)$$

Application of inverse Laplace transform to Eq. (4.13) produces

$$\mathcal{L}^{-1}[\text{Eq. (4.13)}] = \hat{N} \left[1 + 2\mathcal{V} \ln \left(\frac{2\pi\hat{N}}{L} \right) \right]. \quad (4.14)$$

$\langle \mathbf{R}^2 \rangle$ can be obtained now as¹⁶

$$\langle \mathbf{R}^2 \rangle = - \frac{2d(\partial/\partial \mathbf{p}^2) \mathcal{L}^{-1}[G(\mathbf{p}, s)]}{\mathcal{L}^{-1}[G(\mathbf{p}, s)]} \Big|_{\mathbf{p}=0}. \quad (4.15)$$

Combining Eqs. (4.5), (4.14), and (4.15) we obtain

$$\begin{aligned} \langle \mathbf{R}^2 \rangle &= \hat{N} \left[1 + 2\mathcal{V} \ln \left(\frac{2\pi\hat{N}}{L} \right) \right] \left[1 - \mathcal{V} \left[1 + \ln \left(\frac{2\pi\hat{N}}{L} \right) \right] \right] \\ &\simeq \hat{N}(1 - \mathcal{V}) \left[1 + \mathcal{V} \ln \left(\frac{2\pi\hat{N}}{L} \right) \right] + O(\mathcal{V}^2). \end{aligned} \quad (4.16)$$

For $\mathcal{V} = \mathcal{V}^* = \epsilon/8$ we obtain back the well known result¹⁶

$$\langle \mathbf{R}^2 \rangle = \hat{N} \left(\frac{2\pi\hat{N}}{L} \right)^{\epsilon/8} \left(1 - \frac{\epsilon}{8} \right). \quad (4.17)$$

The above result was obtained in a system of units where $d/l=1$. Going to usual system of units, Eq. (4.17) may be also rewritten as

$$\langle \mathbf{R}^2 \rangle = \hat{N} l \left(\frac{2\pi\hat{N}}{L} \right)^{\epsilon/8} \left(1 - \frac{\epsilon}{8} \right). \quad (4.18)$$

Using Eq. (4.12) and the results of Sec. III, we would like to rewrite Eq. (4.18) in the following equivalent form:

$$\langle \mathbf{R}^2 \rangle = \hat{N}^{1+(\epsilon/8)} m^{(\epsilon/8)-1} \text{const.} \quad (4.19)$$

Using Eqs. (4.3) and (4.9), we can now write for the propagator $\hat{G}(\mathbf{p}, s)$ the following result:

$$\begin{aligned} \hat{G}(\mathbf{p}, s) &= \frac{2d(s+m)}{(\mathbf{p}^2/2) + M^2} - \frac{\mathcal{V}[2d(s+m)]^3}{[(\mathbf{p}^2/2) + M^2]^2} \\ &\times M^2 \left[\ln \left(\frac{M^2 L^2}{2\pi} \right) - \psi(2) \right]. \end{aligned} \quad (4.20)$$

From here we obtain

$$\begin{aligned} -2d \frac{\partial}{\partial \mathbf{p}^2} \hat{G}(\mathbf{p}, s) \Big|_{\mathbf{p}=0} &= \frac{2d^2(s+m)}{M^4} \\ &- \frac{2d\mathcal{V}}{M^4} [2d(s+m)]^3 \left[\ln \left(\frac{M^2 L^2}{2\pi} \right) - \psi(2) \right], \end{aligned} \quad (4.21)$$

where, as before, $M^2 = d(s^2 - m^2)$. The last result can be also rewritten in the form

$$\begin{aligned} \text{Eq. (4.21)} = & \frac{2}{(s-m)^2(s+m)} - \frac{\mathcal{V}[4d]^2(s+m)}{(s-m)^2} \\ & \times \left[\ln \left[\frac{(s-m)}{2\pi} \sqrt{dL} \right] \right. \\ & \left. + \ln \left[\frac{(s+m)}{2\pi} \sqrt{dL} \right] - \psi(2) \right]. \quad (4.22) \end{aligned}$$

Applying the inverse Laplace transform to the first term in the right-hand side of Eq. (4.22) produces

$$\begin{aligned} \mathcal{L}^{-1} \left[\frac{2}{(s-m)^2(s+m)} \right] = & \left[\frac{-2}{(2m)^2} + \frac{2\hat{N}}{2m} \right] \\ & \times e^{m\hat{N}} + \frac{2}{(2m)^2} e^{-m\hat{N}}. \quad (4.23) \end{aligned}$$

When $\mathcal{V} = 0$, use of Eqs. (4.11), (4.15), and (4.23) produces back the result, Eq. (3.20), as expected. Applying the inverse Laplace transform to the second term in the right-hand-side, Eq. (4.22) requires more work along the lines similar of that presented in the Appendix. In particular, we obtain

$$\mathcal{L}^{-1} \left[\frac{s+m}{(s-m)^2} \right] = 2m\hat{N}e^{m\hat{N}}, \quad (4.24)$$

$$\mathcal{L}^{-1} \left[\frac{1}{(s-m)^2} \ln(s-m) \right] = e^{m\hat{N}}\hat{N}[\psi(2) - \ln \hat{N}], \quad (4.25)$$

$$\mathcal{L}^{-1} \left[\frac{1}{(s-m)^2} \ln(s+m) \right] = -e^{-m\hat{N}}\hat{N} \ln 2m, \quad (4.26)$$

etc. Using Eqs. (4.24)–(4.26) and the results of the Appendix, we are ready to write the inverse Laplace transform of the right-hand side of Eq. (4.22). It is given by

$$\begin{aligned} \mathcal{L}^{-1}[\text{Eq. (4.22)}] = & \left[\frac{-2}{(2m)^2} + \frac{2\hat{N}}{2m} \right] e^{m\hat{N}} + \frac{2}{(2m)^2} e^{-m\hat{N}} \\ & + \frac{2u\hat{N}}{m} e^{m\hat{N}} \ln(2m\hat{N}). \quad (4.27) \end{aligned}$$

In arriving at Eq. (4.27) we have used Eq. (4.12). Use of Eqs. (4.11), (4.15), and (4.27) allows us to write for $\langle \mathbf{R}^2 \rangle$ the following result:

$$\begin{aligned} \langle \mathbf{R}^2 \rangle \simeq & \frac{\hat{N}}{2m} [1 + u \ln(2m\hat{N})] (1-u) \\ & - \frac{(1-u)}{(2m)^2} (1 - e^{-2m\hat{N}}) [1 - u \ln(2m\hat{N})] \quad (4.28) \end{aligned}$$

to be compared with Eq. (4.16). For nonzero m and for $u = u^*(\epsilon/8)$, the last result can be rewritten, as before, in the following scaling form:

$$\begin{aligned} \langle \mathbf{R}^2 \rangle = & \left(1 - \frac{\epsilon}{8} \right) \left[\frac{\hat{N}}{2m} (2m\hat{N})^{\epsilon/8} \right. \\ & \left. - (2m\hat{N})^{-\epsilon/8} \frac{1}{(2m)^2} (1 - e^{-2m\hat{N}}) \right]. \quad (4.29) \end{aligned}$$

This result holds only for $\hat{N} \rightarrow \infty$ and $m > 0$. For $\hat{N}$ finite (or $m \rightarrow 0^+$), the scaling result given by Eq. (4.29) is not correct. In this limit $u \rightarrow 0$ [because $u = \mathcal{V}(4md)^2$] and we recover back the rigid rod result: $\langle \mathbf{R}^2 \rangle \propto \hat{N}^2$. Because of this observation, it is essential to study the crossover problem in more detail. This is accomplished in the next section.

V. THE PROBLEM OF CROSSOVER

As in the fully flexible case, it is of interest to consider not only the scaling limits of observable quantities but also the intermediate temperature-dependent situations. It is believed,² that the transition from the good solvent to the poor solvent regime in the case of fully flexible *nonrandom* polymers is solely due to the variations in temperature. The strength of the excluded volume interactions is believed to be weakly temperature dependent, as it is explained in Ref. 1, while the phenomenological parameter L entering into all renormalized quantities of interest² is also weakly temperature dependent. The temperature dependence of the excluded volume interactions is known in principle¹ and is responsible for the crossover effects while the temperature dependence of L is unknown and enters into calculations only indirectly through the scaling variables (to be discussed below), which are some algebraic combinations of both excluded volume interaction parameter $\mathcal{V}$ and L . In cases of semiflexible polymers, the length L is no longer a parameter of the theory; it is fixed by the consistency condition, Eq. (4.12). We also have learned that the coupling constant u , see Eq. (4.11), is connected with the inverse rigidity parameter m . Because of this observation, the crossover for the semiflexible polymers is much more interesting to study. Indeed, the parameter m can be temperature dependent, as it was explained in Sec. III. This then produces the coupling constant u which is also temperature dependent. On the other hand, for fixed temperature it may change because of the changes in the ionic environment (in cases of polyelectrolyte chains) as it was described in Sec. I. In both cases we shall assume that the way the parameter m changes with temperature T and with ionic strength I (for fixed temperature) is known explicitly, i.e., function $m = m(T, I)$ is assumed to be given. In this case the crossover functions, to be introduced below, could be a very complicated functions of external conditions.

To begin our crossover calculations, following Ref. 16, we observe that with help of some length λ the dimensional analysis implies the following result for $\langle \mathbf{R}^2 \rangle$:

$$m \langle \mathbf{R}^2 \rangle = \lambda \mathcal{F} [u, (\hat{N}/\lambda), m\lambda], \quad (5.1)$$

where $\mathcal{F}$ is some, yet unknown, dimensionless scaling function. Notice, that although the coupling constant u is dimensionless it, nevertheless, depends on m . From the previous discussion it follows that we have two options:

(a) to keep u as an adjustable parameter of the theory, (b) to account for the explicit dependence of u upon m [see, e.g., Eq. (4.11)]. Fortunately, in the present case both situations can be treated simultaneously. In Ref. 16 we have demonstrated that the scaling function $\mathcal{F}$ introduced in Eq. (5.1) satisfy the following renormalization group equation:

$$\left[L \frac{\partial}{\partial L} + \beta(u) \frac{\partial}{\partial u} + \gamma(u) \hat{N} \frac{\partial}{\partial \hat{N}} \right] \mathcal{F}(u, \hat{N}, L, \epsilon) = 0, \quad (5.2)$$

where

$$\beta(u) = \frac{\epsilon}{2} u \left(1 - \frac{8}{\epsilon} u \right) + O(u^3) \quad (5.3)$$

and

$$\gamma(u) = u + O(u^2). \quad (5.4)$$

Because of Eq. (4.12), it is also possible to rewrite Eq. (5.2) in terms of m rather than L . We shall do it, however, somewhat later. In the meantime, we would like to notice a formal mathematical similarity between Eq. (5.2) and Eq. (3.15) of our previous work, Ref. 15. This similarity enables us to write without delay a formal solution of Eq. (5.2) in the following way:

$$\mathcal{F}(u, \hat{N}, L = m_0^{-1}) = \mathcal{F}[u(l), \hat{N}(l), m(l) = m_0 e^{-l}], \quad (5.5)$$

where $u(l)$ and $\hat{N}(l)$ satisfy the following differential equations:

$$\frac{du}{dl} = \beta[u(l)], \quad (5.6)$$

$$\frac{d\hat{N}}{dl} = \hat{N}(l) u(l), \quad (5.7)$$

with initial condition $u(l=0) = u$, $\hat{N}(l=0) = \hat{N}$. Equation (5.5) has the following meaning. On the left-hand side of this equation we have a scaling function which is written for some fixed inverse rigidity constant m_0 which, by virtue of Eq. (4.12), defines the strength of the excluded volume parameter u at some scale L . On the right-hand side of Eq. (5.5) we have the same scaling function but written for different rigidity $m_0^{-1} e^l$, where l is determined by the equation: $l = \ln(\lambda/L)$.

Following Ref. 15, and using Eq. (4.28) we can formally write

$$m_0 \langle \mathbf{R}^2 \rangle = [1 - u(l)] \frac{1}{4m(l)} \{ N(l) [1 + u(l) \ln N(l)] - [1 - u(l) \ln N(l)] (1 - e^{-N(l)}) \}, \quad (5.8)$$

where we have introduced a new variable $N(l) = 2m(l) \hat{N}(l)$. Define now the matching point l^* via condition,

$$N(l^*) = 1. \quad (5.9)$$

The meaning of this matching point can be understood, in part, by considering a similar situation described in Ref. 15. In case if Eq. (5.9) holds the logarithmic corrections

present in Eq. (5.8) vanish which is *formally* equivalent to the statement that under the matching condition, Eq. (5.9), the effects associated with the excluded volume interactions are unimportant. This statement is somewhat misleading, though, as it will become apparent upon reading.

For the *fixed* nonzero m_0 a great deal of simplifications are possible in Eq. (5.8) thus bringing it into the form

$$\langle \mathbf{R}^2 \rangle = \frac{1}{(2m_0)^2} [1 - u(l^*)] e^{l^* - 1}. \quad (5.10)$$

This simplification breaks down when the limit $m_0 \rightarrow 0$ is considered. For the time being, we shall assume that m_0 is *strictly* greater than zero. In this case we need to find e^{l^*} and $u(l^*)$ in order to complete our calculations of $\langle \mathbf{R}^2 \rangle$. For this purpose we need to replace the equation for $\hat{N}(l)$, Eq. (5.7), by that for $N(l)$. Combining Eq. (5.7) with the definition of $N(l)$ we obtain

$$\frac{dN(l)}{dl} = N(l) [u(l) - 1]. \quad (5.11)$$

Combined use of Eqs. (5.3) and (5.6) produces the following result for $u(l)$:

$$u(l) = u^* \left[1 + \frac{u^* - u}{u} e^{-4u^* l} \right]^{-1}, \quad (5.12)$$

where $u^* = \epsilon/8$. Following closely the results of Ref. 15, we would like to investigate the stability of the fixed point $u^* = \epsilon/8$. Combining Eqs. (5.3) and (5.6) and using $u = u^* + \delta u$ we obtain

$$\frac{d\delta u}{dl} = -\frac{\epsilon}{2} \delta u \equiv -4u^* \delta u. \quad (5.13)$$

From here $\delta u = \delta u_0 e^{-4u^* l} \equiv g_u e^{-4u^* l}$, where the last equality defines Wegner–Riedel²² scaling variable g_u . As in Ref. 15, the amplitude g_u can be identified with $(u^* - u)/u$ by comparing Eqs. (5.12) and (5.13). In view of such identification, Eq. (5.12) can be rewritten as

$$u(l) = u^* [1 + g_u(l)]^{-1}, \quad (5.14)$$

where the scaling field $g_u(l)$ is given by

$$g_u(l) = \frac{u^* - u}{u} e^{-4u^* l}. \quad (5.15)$$

It is useful to introduce another Wegner–Riedel scaling field $g_N(l)$ by considering stability of $N(l)$ at the fixed point for u . We have with help of Eq. (5.11) the following result:

$$\frac{d\delta N}{dl} = \delta N (-1 + u^*). \quad (5.16)$$

This produces

$$\delta N(l) = g_N e^{(-1 + u^*) l}. \quad (5.17)$$

In order to find explicitly g_N consider once again Eqs. (5.6) and (5.7). Combining these equations we obtain

$$\frac{d\hat{N}(l)}{du(l)} = \frac{\hat{N}(l)u(l)}{(\epsilon/2)u(l)[1 - (u(l)/u^*)]} \quad (5.18)$$

From here it follows, that

$$\hat{N}(l) = \hat{N} \left[\frac{u^* - u(l)}{u^* - u} \right]^{1/4} \quad (5.19)$$

or, in terms of $N(l)$,

$$N(l) = 2m_0 e^{-l\hat{N}} \left[\frac{u^* - u(l)}{u^* - u} \right]^{1/4}. \quad (5.20)$$

Use of Eqs. (5.14) and (5.15) allows us to rewrite Eq. (5.20) in the form

$$N(l) = 2m_0 \hat{N} e^{(-1+u^*)l} \left(\frac{u}{u^*} \right)^{1/4} [1 + g_u(l)]^{1/4}. \quad (5.21)$$

Combining Eqs. (5.17) and (5.21) produces

$$N(l) = g_N(l) [1 + g_u(l)]^{1/4}, \quad (5.22)$$

where we have introduced scaling field $g_N(l)$ via

$$g_N(l) \equiv g_N e^{(-1+u^*)l} = 2m_0 \hat{N} \left(\frac{u}{u^*} \right)^{1/4} e^{(-1+u^*)l}. \quad (5.23)$$

Combining Eqs. (5.9) and (5.22) produces equation for the matching point l^* ,

$$g_N(l^*) [1 + g_u(l^*)]^{1/4} = 1. \quad (5.24)$$

Using results of Ref. 15, it is convenient to rewrite Eq. (5.24) in the following equivalent form:

$$e^{l^*} = g_N e^{(\epsilon/8)l^*} [1 + g_u e^{(\epsilon/2)l^*}]^{1/4}. \quad (5.25)$$

In order to solve Eq. (5.25) we use the method of iteration. For the first iteration step we obtain $e^{l^*} \sim g_N^{1+(\epsilon/8)}$, while for the second we get

$$e^{l^*} = g_N^{1+(\epsilon/8)} [1 + g_u g_N^{-\epsilon/2}]^{1/4} \equiv g_N^{1+(\epsilon/8)} \Psi(x), \quad (5.26)$$

where the crossover function $\Psi(x)$ is given by

$$\Psi(x) = (1+x)^{1/4} \quad (5.27)$$

and $x = g_u g_N^{-\epsilon/2}$. The result, Eq. (5.26), is useful to compare with Eqs. (4.15)–(4.17) of Ref. 15.

In order to write the final form of $\langle R^2 \rangle$ given by Eq. (5.10), we would like to rewrite Eq. (5.14) in the following form:

$$u(l) = \frac{u^*}{1+x}. \quad (5.28)$$

In arriving at this result Eqs. (5.26) and (5.27) were used. Combining Eqs. (5.26), (5.28), and (5.10) we obtain for $\langle R^2 \rangle$ the correct Wegner–Riedel crossover result

$$\langle R^2 \rangle = \frac{g_N^{1+(\epsilon/8)}}{(2m_0)^2} \Psi(x) \left[1 - \frac{u^*}{1+x} \right] e^{-1}. \quad (5.29)$$

To understand better what it actually means, consider the case of three dimensions: $\epsilon = 1$. In this situation we obtain

$$x = \frac{u^* - u}{u} \left[2m_0 \hat{N} \left(\frac{u}{u^*} \right)^{1/4} \right]^{-1/2}. \quad (5.30)$$

For the fixed u and $\hat{N}$ the crossover variable x depends only on the inverse rigidity parameter m_0 . For m_0 nonzero and $\hat{N}$ large $x \ll 1$ and we obtain as a leading approximation

$$\langle R^2 \rangle \propto g_N^{1+(1/8)} \propto \hat{N}^{1+(1/8)}, \quad (5.31)$$

i.e., usual excluded volume-type behavior. Now let $m_0 \rightarrow 0^+$, then $x \gg 1$ and we obtain, using Eq. (5.29), the following leading result (for fixed $\hat{N}$):

$$\langle R^2 \rangle \propto \frac{\hat{N}}{m_0} \rightarrow \infty, \quad (5.32)$$

which brings us back to Eq. (5.8) [see the discussion following Eq. (5.10)]. In order to further analyze this equation, several observations are of some help. First, we would like to notice that the effective coupling constant $u(l)$ given in Eq. (5.28) tends to zero for $l = l^*$ and $m_0 \rightarrow 0$, see Eq. (5.30). Therefore, in this case the matching point condition $N(l^*) = 1$ loses its importance: because in this regime the logarithmic terms in Eq. (5.8) are going to disappear anyway. But when $u(l^*) \rightarrow 0$ (or there is no logarithmic terms) we have the situation when the excluded volume effects lose their significance and we are left with the free chain situation. Moreover, let, for instance, instead of matching condition, Eq. (5.9), we would choose

$$N(l^*) = 2m_1 \hat{N} \leq 1 \quad (5.33)$$

under conditions that $m_0 \rightarrow 0^+$. Then, instead of Eqs. (5.24) and (5.25), we would have

$$e^{l^*} = \frac{m_0}{m_1} \left(\frac{u}{u^*} \right)^{1/4} [1 + g_u e^{-(\epsilon/2)l^*}]^{1/4} e^{(\epsilon/8)l^*} \quad (5.34)$$

and instead of g_N in Eq. (5.26) we would have $(m_0/m_1) \times (u/u^*)^{1/4}$. When $m_0 \rightarrow 0$ this would lead us anyway to the result $u(l^*) \rightarrow 0$, as expected. In this regime, using Eq. (5.8), we obtain

$$\langle R^2 \rangle = \frac{m_1 \hat{N}}{2m_0 m(l^*)} - \frac{1}{4m_0 m(l^*)} (1 - e^{-2m_1 \hat{N}}). \quad (5.35)$$

But $m(l^*) = m_0 e^{-l^*} \approx m_1 [(u^* - u)/u^*]^{1/4} \approx m_1 [1 - (u/4u^*)]$ in view of Eqs. (5.15) and (5.34). Let $m_1 < m_0$ (e.g., let $m_1 = m_0 [1 - (u/4u^*)]$), then expanding the exponent in Eq. (5.35) we obtain back known result: $\langle R^2 \rangle = \hat{N}^2$. As we have demonstrated, in all cases the rigidity effects play a crucial role in the crossover effects. This observation is especially important in case of polyelectrolytes where the effective rigidity changes with ionic strength for the fixed temperature.

VI. DISCUSSION AND THE SUMMARY

In the previous sections we have developed a new renormalization group treatment of semiflexible polymers with excluded volume. Among the most important results we have been able to obtain, we would like to mention:

(1) The renormalization group analysis for the semiflexible polymers technically does not require new elements (except more complicated inverse Laplace transforms) as compared with the fully flexible case.¹⁶

(2) In the fully flexible case the final result of calculations contained some *unspecified* parameter L which has dimensionality of length. Our calculations *completely* remove the arbitrariness of choice of L and connect it with microscopically calculable quantities such as inverse rigidity (related, e.g., to *trans-gauche* bending energy).

(3) Our crossover calculations have demonstrated a crucial role of rigidity in the crossover from the rigid rod to the random coil conformations. This is especially plausible for polyelectrolytes where the effective rigidity (at fixed temperature) is solely responsible for rigid rod-to-random coil transition. A rather complicated crossover behavior observed for polyelectrolytes^{6,11} now can be explained if the functional dependence of the rigidity on the ionic strength I of solution is known. More detailed calculations of this dependence are going to be performed in the future.

In spite of the above achievements, a whole number of problems remains to be solved *even for the single chain case*. First, in Sec. III we have provided plausible arguments that $1 + 1$ dimensional Dirac propagator, Eq. (3.16), can be used in $d + 1$ dimensional calculations. To prove that this is indeed the case, we have indicated earlier⁹ that instead of Ising model, used in Sec. III, we have to use one dimensional $2d$ -component Potts model. The calculations for this case have been already performed¹⁰ for both homopolymer and random copolymer cases (see Sec. I). In the last case it was possible to obtain the results without the recourse to the replica trick methods. The obtained results need to be further extended to account for the effects of boundaries and interfaces. This might require us to provide an explicit functional integral for Dirac's propagator.⁹ We would also like to have this propagator in order to study the problems related to elasticity of rubbers and crosslinked ionic gels. These and other projects are left for future work.

Note added: After this work was completed, we became aware of the recent experimental results²³ regarding the conformation properties of semiflexible polymers in dilute polymer solutions. The authors of Ref. 23 were motivated by the idea that "to explain dilute solution properties of linear polymers over a broad range of chain stiffness or chain length *even a flexible polymer should be treated as semiflexible*." To this purpose the existing Yamakawa-Stockmayer (Y-S) perturbation theory²⁴ for the wormlike chains with excluded volume was combined with the Domb-Barret (D-B) expansion²⁵ for the fully flexible polymers in order to explain the observed crossover of the radius of gyration from good to theta solvent regime.

For the case of hexyl isocyanate (PHIC) in hexane, the observed crossover is rather sharp and *cannot* be explained by the combined Y-S and D-B expansions. In the same time, for the polymer compound PIDP in N methyl-2 pyrrolidone (NMP) the observed crossover is rather smooth and *can* be described in terms of the above combined expansions.

The theory developed in Sec. V provides a correct choice of scaling variables related to Y-S and D-B expansions in an obvious way.² The ultimate test of the developed crossover theory is expected to provide the most ac-

curate data for the persistence length which, in turn, could be used in order to study the role of solvent effects on the *trans-gauche* bending energies (the last ones, in principle, can be also computed quantum mechanically).

APPENDIX: CALCULATIONS RELATED TO EQ. (4.10)

Denote $\mathcal{L}^{-1}$ the operation of taking the inverse Laplace transform. Then we have, e.g.,

$$\mathcal{L}^{-1}\left[\frac{1}{s-m}\right] = e^{mN}. \quad (\text{A1})$$

In order to compute $\mathcal{L}^{-1}[(s+m)^2/(s-m)]$ consider the following auxiliary inverse Laplace transform:

$$\mathcal{L}^{-1}\left[\frac{e^{-as}}{s-m}\right] = \begin{cases} 0 & \text{for } N < a \\ e^{m(N-a)} & \text{for } N > a. \end{cases} \quad (\text{A2})$$

From here

$$\mathcal{L}^{-1}\left[\frac{s}{s-m}\right] = -\left.\frac{\partial}{\partial a} e^{m(N-a)}\right|_{a=0} = m e^{mN}. \quad (\text{A3})$$

Analogously,

$$\mathcal{L}^{-1}\left[\frac{s^2}{s-m}\right] = m^2 e^{mN}, \text{ etc.} \quad (\text{A4})$$

So that finally we obtain

$$\mathcal{L}^{-1}\left[\frac{(s+m)^2}{s-m}\right] = 4m^2 e^{mN}. \quad (\text{A5})$$

Consider now $\mathcal{L}^{-1}[(1/(s-m)) \ln(s \pm m)]$. In order to calculate this transform consider the auxiliary problem of calculation of $\mathcal{L}^{-1}[\frac{1}{s} \ln s]$. Taking into account that¹⁹

$$\ln\left(\frac{s}{1}\right) = -\int_0^\infty \frac{dx}{x} (e^{-sx} - e^{-x}) \quad (\text{A6})$$

and using the fact that

$$\mathcal{L}^{-1}\left[\frac{1}{s} e^{-sx}\right] = \theta(N-x), \quad (\text{A7})$$

we obtain

$$\begin{aligned} \mathcal{L}^{-1}\left[\frac{1}{s} \ln s\right] &= -\int_0^\infty \frac{dx}{x} [\theta(N-x) - e^{-x}] \\ &= -[\ln N - \ln \hat{e} - e^{-x} \ln x]_{\hat{e}}^\infty \\ &\quad - \int_0^\infty dx e^{-x} \ln x \\ &= \psi(1) - \ln N. \end{aligned} \quad (\text{A8})$$

Here $\hat{e} = \lim_{x \rightarrow 0^+} x$ and, after the integration by parts was performed, we have used the known result: $\psi(1) = \int_0^\infty dx e^{-x} \ln x$. Consider now $\mathcal{L}^{-1}[(1/(s-m)) \ln(s-m)]$. Substitution of $s' = s + m$ brings us back to Eq. (A8) so that we can write without delay

$$\mathcal{L}^{-1}\left[\frac{1}{s-m} \ln(s-m)\right] = e^{mN} [\psi(1) - \ln N].$$

Calculation of $\mathcal{L}^{-1}[(1/(s-m))\ln(s+m)]$ is somewhat more involved. We have

$$\begin{aligned}\mathcal{L}^{-1}\left[\frac{1}{s-m}\ln(s+m)\right] \\ = -e^{mN}\left[\int_0^\infty \frac{dx}{x}(e^{-2mx}\theta(N-x) - e^{-x})\right] \\ = -e^{mN}\left[\int_0^N \frac{dx}{x}(e^{-2mx} - e^{-x})\right].\end{aligned}\quad (\text{A10})$$

This requires us to consider the calculation of

$$\begin{aligned}I_1 &= \int_0^N \frac{dx}{x} e^{-2mx} - \int_0^\infty \frac{dx}{x} e^{-x} \\ &= e^{-2mx} \ln x \Big|_0^N + \ln \hat{e} - \psi(1) + 2m \int_0^N dx e^{-2mx} \ln x,\end{aligned}\quad (\text{A11})$$

but

$$\begin{aligned}I_2 &= \int_0^N dx e^{-2mx} \ln x \\ &= \int_0^\infty dx e^{-2mx} \ln x - \int_N^\infty dx e^{-2mx} \ln x \\ &= \frac{1}{2m}[\psi(1) - \ln 2m] - \left[\frac{1}{-2m}e^{-2mx}\right]_N^\infty \\ &\quad + \frac{1}{2m} \int_N^\infty \frac{dx}{x} e^{-2mx}.\end{aligned}\quad (\text{A12})$$

Taking into account that

$$\int_N^\infty \frac{dx}{x} e^{-2mx} = \int_{2mN}^\infty \frac{dy}{y} e^{-y} \simeq \frac{e^{-2mN}}{2mN} \left[1 - \frac{1}{2mN} + \dots\right]\quad (\text{A13})$$

and collecting all relevant terms we obtain

$$I_1 \simeq -\ln 2m - e^{-2mN} \left[1 - \frac{1}{2mN} + \dots\right].\quad (\text{A14})$$

For m finite and $N \rightarrow \infty$ the last term in Eq. (A14) can be ignored. But when $m \rightarrow 0^+$ (the rigid rod limit), it cannot be ignored, in principle. Taking into account that in this limit the coupling constant vanishes allows us to ignore this term, nevertheless. Calculation of $\mathcal{L}^{-1}\{(s+m)^2/(s-m)\ln(s \pm m)\}$ can now be performed based on the results obtained above and taking into account that

$$\mathcal{L}^{-1}\left[\frac{e^{-(x+a)s}}{s-m}\right] = e^{m(N-x-a)}\theta(N-x-a) \quad (\text{A15})$$

which need to be compared with Eq. (A2). This concludes our calculations related to Eq. (4.10).

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